

AMI

Moist O2 inhalation@6-8 L/min

S/R diet.(if diabetic+ Htn then s/r Diabetic diet)

Tab. Sorbitrate(5mg) 1 tab sublingual stat, spit the tab out once chest pain relieved.

Tab. Aspirin (75mg) 4tab stat then 1tab ODPC× cont.

Tab. Clopidogrel(75mg) 4 tab stat then 1 tab ODPC× cont

Tab. Atorvastatin (40mg) 2 tab stat then 1 tab ODHS× cont.

Tab. Nitrocontin(2.6mg) 1 tab BD× Cont. (8am and 8 pm) (Only if SBP >90mmHg)

Inj. Ondem(4mg) Iv TID× cont.

Inj. Pantoprazole (40mg) iv stat then iv odac× cont.

Cardiac Remodelling prevention:

Tab. Ramipril(2.5mg) 1 tab OD× cont.

Tab. Metoprolol(25mg) ½ tab OD × Cont.

Tab. Alprazolam (0.25) or Clonazepam(0.5mg) 1 tab ODHS× cont.

Syp. Lactulose – 15ml ODHS× Cont.

Inj Enoxaparin(40) S/C OD× 7days.

Monitor Vitals.

INVESTIGATION:

Trop T

CXR- Pa view

Urgent Na,K,ur,Cr (exclude CKD)

2D ECHO

CARDIO REFER MUST.

CKD

BP,CBG – Must

ADVICE

1. S/R Renal Diabetic diet. (a/c to comorbidities) + Moist O2 if hypoxic + Railed cot/Water bed if Uremic Encephalopathy present.

2. Fluid restriction, Intake= output+ 10 ml/kg

3. Inj. Pipzo(4.5g) IV stat then 2.25g IV TID× cont(APST)

4. Inj. Lasix(20mg) IV bd × cont.

5 . Inj Ranitidine (50) IV BDAC × cont.

6. Inj. Ondansetron(4) IV stat and IV TID ×cont.

7. Tab. Ca + Vit D3 (500mg + 250 iu) 1 tab ODPC × cont.

8. Cap. Rocaltrol (0.25 ug) 1 cap od × cont.

9. Tab. IFA 1 tab OD × cont.

10. Tab. NaHCO₃ (NODOSIS) 500mg 1 tab BD× cont.

11. DRUG A/C TO COMORBIDITIES.

Syr. LACTULOSE if stool not passed.

PRBC requisition a/c to hb level <7gm/dl.

Investigation:

Urgent Na, k, ur, cr, CBC

E+C

Urine R/E, M/E

CXR – PA (to R/o pneumonia, CHF)

USG W/A(no need if k/C/o CKD)

Serology

12lead ECG.

S. Ca²⁺, PO₄, CRP, VitD₃

Nephro refer.

HISTORY

Known CKD or not. Stage?

On HD or not? How many HD per wk.

Serology? USG? Recent Na, k, ur, cr, Hb.

Any cause- HTN, DIABETIC, HYPOTHYROIDISM, PSGN, HUS(h/o diarrhoea, hematuria) .

Dr. Subhajit Mondal, GM U-3, NRSMCH.

SARI

Mild SARI, no comorbidities, not suggestive of Covid

Moist O₂ @2-4 L/min with face mask.

Inj Ceftriaxone (1g) iv bd× cont.

Inj Pantoprazole(40mg) IV stat then odac× cont.

Inj Ondem(4) IV stat then IV BD/TDS× Cont.

Tab. Vit C (500) 1 tab OD× Cont

Tab. Zn(50) 1 tab OD× cont

Treat other comorbidities, if present.

NB - If patient can take food orally, vital stable then no IV fluid required.

If severe pallor or heart failure, ckd, IVF is contraindicated.

For Diabetic, Give NS/ RI

Mild SARI, CXR suspicious of COVID, give

Tab. DOXYCYCLINE(100mg) 1 tab BD× 7days.

Tab. Ivermectin (12 mg) 1tab OD × 5days.

Pan

Ondem

If patient is diabetic, Total Count High

Can be started with Inj. Pipzo(4.5 g) IV stat then IV TDS× cont (APST)

MODERATE SARI:

DOXY

IVERMECTIN

Inj Dexamethasone(0.1-0.2 mg/kg) IV in divided doses.

Inj. LMW Heparin(40) S/C OD× cont. (to overcome risk of thromboembolic event)

Tab Vit C (500) 1 tab OD× cont

Tab. Zn(50) 1 tab OD× cont.

#Severe SARI

Fluid

Moist O2 @ 6-8 ltr with face mask, to increase if spo2 falls.

RT/CATHETER if required.

Inj. Pipzo(4.5) IV STAT then IV TDS× cont (APST)

Tab. Doxycycline(100) 1 tab bd

Tab. Ivermectin(12) 1 tab OD

Inj. Dexamethasone(0.2-0.4mg/Kg BW) IV divided doses.

Inj LMW Heparin (40- 60 mg) s/C BD× Cont. If CKD, then UFH may be considered.

Tab. PCM(650) 1 tab SOS.

Vit C , Zn

Pan Ondem

Comorbidities if any treat accordingly.

Management of Heart Failure

History and Clinical Examination :

B/L basal crepts, S3 gallop, cold clammy extremity with low volume pulse, tachycardia, pulsus alternans, H/O Exertional SOB with PND , pink frothy sputum- s/o LVF.

Raised JVP, B/L pedal edema, soft tender HEPATOMEGALY points to RVF.

Search for etiology: AMI, Valvular heart disease, Lungs disease(COPD) , Severe anemia, AV block.

Search for precipitating event: LRTI, Stressful condition. (Like DKA)

INVESTIGATIONS:

CBG, CBC, LFT, RFT, LIPID PROFILE, Chest XRay: PA view(digital) , 12 LEAD ECG, Trop T, 2D Echo, USG w/a to r/o any other association(cirrhosis,CKD) .

Other investigation- (not routinely)

Thyroid function test(if suspected)- fT4,TSH., Urine R/M if diabetic, Sputum for Gram stain, c/s ,fungal stain , ZN stain, Cbnaat if s/o LRTI, PT,aPTT, INR if patient is already on anticoagulant therapy due to valvular heart disease)

Treatment:

Moist O2 inhalation at required rate. (Depending on level of desaturation)

Propped up posture

Fluid restriction< 1.5 l/day

I/O chart

Diet: Salt restricted Diet(add as per Comorbidity)

Inj. Lasix(20mg) IV stat then IV TDS× cont.

Tab. Aldactone (Spironolactone) 25 mg 1 tab OD× cont

Tab. Carvedilol (3.125mg) 1 tab BD× cont.

Use vasopressor if patient is in cardiogenic shock[Noradrenaline- 1 amp(4mg) in 500 ml DNS in microdrip set@10 microdrop/ min.--→ may be escalated to Dobutamine if BP still falling @ 6-8 micro drop/min in NS.]

Syp. Lactulose- 15 ml ODHS× cont, P-enema SOS.

ANTIBIOTICS: Inj. Ceftriaxone (1g) IV stat then BD× Cont(AST)

[If AMI is precipitating event add Tab. Aspirin (75) + Clopidogrel (75) + Atorvastatin (40mg). Don't give nitrates,]

Tab Ramipril(2.5mg) BD / Tab. Telmisertan(40) OD if BP remains high. (> 160/100)

Inj. Pantoprazole (40) IV ODAC × Cont.

Tab. Domperidone (10) 1 tab TDS × Cont.

Treat COMORBIDITY(HTN, DM, THYROID D/O , COPD, Severe anemia if present).

Thank you. 😊

Subhajit Mondal(Unit III, GM)

Management of Hyperkalemia($S.K^{+} > 5.5 \text{ mEq/L}$)

Treat the cause. (CKD- DIALYSIS, DRUG INDUCED like K sparing agent, ACEI OR ARB etc) , 1° hypoaldosteronism/ Addison's.

Inj Regular Insulin 10unit IV stat with 100ml 25 % Dextrose solution, if CBG<250 mg/dl. Otherwise no dextrose.

Salbutamol MDI (LEVOLIN) 2puff 6hrly

K bind Sachet(Ca polystyrene Sulfonate) BD (each sachet to be dissolved in 200 ml of normal water, sorbilin may be added to prevent colonic necrosis)

10% calcium gluconate- 1 amp(10ml or 93mg of elementary Ca^{2+})slowly IV over 10 min stat. Baseline ECG to be done before IV.

Inj. Lasix(10mg) IV bdx cont. with oral salt intake + IVF NS 12 hrly (if normovolemic)

Patiromer. N/A in market.

Management of Hyponatremia

Acute(within 48 hrs)

To be corrected not more than 8-10mEq/L/ 24 hrs and not more than 18mEq/L in first 48 hrs.

- I) With 3% NaCl- 1 L of 3% NaCl can correct $\text{Na} = (513 - \text{S. Na}) / (\text{TBW} + 1)$
Total Body Water= 60% of Body wt in Male and 50% of Body weight in female.

Similarly, In 0.9% NaCl- 154mEq/L and .45% NaCl- 77mEq/L.

This is called Adrogue-Madias equation. [Only provides Rough estimate]

Rate of correction: not more than 2 mEq/L/hr with 3% NaCl.

- II) Thereafter 0.5 mEq/L/hr with NS.

- III) ORAL SALT INTAKE: When patient becomes conscious.

Send S. Na, K+, 2hr after correction.

Chronic(>48hrs):

Same management but correction will be slower not more than 6-8 mEq/L/24hrs.

Acute UGI Bleeding (PEPTIC ULCER & VARICEAL BLEEDING)

For both cases:

*NPM

*PRBC TRANSFUSION

For Peptic Ulcer:

Infusion Pantoprazole (40mg) @ 8mg/hr in 500ml DNS. If 8 hourly then 2 vial to be dissolved in each bottle of IVF.

For Variceal bleeding:

Inj Octreotide (50 mcg) iv stat (bolus) followed by 50mcg/hr for 2-5 days(avg 3 days)[8 ampoule in 500ml DNS 8 hourly].

OR,

Inj. Terlipressin (2mg) every 4hr till bleeding controlled or for at least 24hrs upto 48 hrs.

[TERLISTAT – comes in 10ml yellow colour ampoule containing 1mg of Terlipressin]

OTHER MEASURES:

If alcoholic, Inj MVI(Multivitamins Injection) in one bottle of IVF OD× Cont.

Inj. Thiamine Hydrochloride (100mg) IV OD× 7 days.

Inj. Ondansetron(4mg) iv tds× cont.(If comes with hemetemesis) otherwise SOS.

Prophylaxis of Hepatic encephalopathy (Not lactulose) . Antibiotic: Inj.

Cefotaxime(2 gm) IV TDS× Cont(AST). Tab. Rifaximin (550mg) 1 tab BD× cont.

[Inj. Tranexamic acid has no role in controlling UGI bleeding.]

Alcohol de-addiction: Tab Acamprosate (333mg) 1 tab BD × 14 days.

Acute Pancreatitis

Investigation: Urgent serum amylase, lipase, CBC,LFT, RFT, Lipid profile, CRP, USG W/A, CECT W/A(if RFT normal). Straight Xray abdomen(erect posture/ left lateral posture) and Chest X-RAY PA.

Treatment:

NPM

Moist O2 SOS

IVF NS:RL(1:1) 4 hourly(depending on volume status and dehydration, urine output)

Inj. Tramadol(50) IV stat then IV TDS × cont(may infuse with IVF ; Max daily dose: 400mg)

Inj. Pantoprazole (40) 1 amp IV stat f/b IV ODAC× cont.

Inj. Ondansetron 4mg IV TDS × cont (max: 16mg/day)

Multivitamin Injection (MVI) in one NS bottle OD × cont.

[If alcoholic, add Inj. Thiamine(100) IV BD× 5 days then Tab. Thiamine 100 mg TDS × 1 week f/b 100 mg OD] for ACUTE ON CHRONIC PANCREATITIS addition of

Panlipase may be considered after acute phase is over. For alcohol withdrawal Tab. Chlordiazepoxide(10) 1 tab TDS × 2 weeks to be given.]

Manage other comorbidities accordingly.

STATUS EPILEPTICUS

What is status?

The duration of seizure activity sufficient to meet the definition of status epilepticus has traditionally been specified as 15–30 min. However, a more practical definition is to consider status epilepticus as a situation in which the duration of seizures prompts the acute use of anticonvulsant therapy. For GCSE(Generalised convulsive status Epilepticus), this is typically when seizures last beyond 5 min.

GCSE is an emergency.

Treatment:

1. Secure airway/ tongue- use mouth gag/ gauze piece between teeth. Moist O2 if saturation drops.

2. Railed Cot, Ryles tube + catheter (obviously not during active convulsion, and only if patient remains disoriented)

3. Inj. Lorazepam(4mg/amp) 1 amp slow iv stat single dose. (0.1 mg/kg)

May be repeated in 5-15 min if seizures recur. [Monitor respiratory drive, as it is notorious for causing respiratory dipression]

4. Inj. Phenytoin (15-20 mg/kg) – 9-10 amp(100mg) in 500ml NS IV bolus over 30min stat.--- > f/b 100mg IV TDS× continue.

5. If tongue bite occurred, then Botroclot solution ,to apply locally (to prevent hemorrhage) and Zytee gel for healing/antiseptic. ENT referral.

6. Treat other comorbidity: CVA, DM, HTN

7. Supportive: Ryles tube feeding 100ml 4 hourly× cont.

Antibiotics: Inj. Ceftriaxone (1g) IV stat and iv BD× Cont(APST),

Inj.Pantoprazole(40) 1 amp iv odac,

Inj.Ondansetron(4mg) IV TDS.

[Remember: Phenytoin is an microsomal enzyme inducer , so dosage of other drugs may be increased including antibiotics.]

Non-convulsive status Epilepticus:

Nonconvulsive status epilepticus (NCSE) refers to a prolonged seizure that manifests primarily as altered mental status as opposed to the dramatic convulsions seen in generalized tonic-clonic status epilepticus. There are 2 main types of NCSE, each of which has a different presentation, cause, and expected outcome. In the first type of NCSE, patients present with confusion or abnormal behavior, suggesting the diagnosis of absence status epilepticus (ASE) or complex partial status epilepticus (CPSE). The second type of NCSE (subtle status epilepticus [SSE]) must be considered in comatose patients who present after a prolonged generalized tonic-clonic seizure and who may have only subtle motor manifestations of a seizure, such as facial or hand twitchings. Whereas the morbidity and mortality in patients with prolonged ASE or CPSE is low, the mortality associated with SSE can exceed 30% if the seizure duration is greater than 60 minutes.

Management of Ischemic CVA:

Counseling to patient party regarding poor prognosis.

Water Bed/ Air mattress/ Frequent change of posture to be encouraged.

If h/o convulsions---→ RAILED COT is a must.

Moist O2- @2-4 L/min with face mask. If distress present.

Diet: S/R(salt restricted) bland diet. Or Renal/Diabetic diet,(according to comorbidities)

RT/Catheterization if patient is unconscious

RT feeding 200ml 4 hourly xcont.

Tab. Aspirin(75) 1 tab odpcxcont.(To be avoided in large infarct) *ASK ME

Tab. Atorvastatin(10) 1tab ODHSxcont.

Antihypertensive(amlo 5 etc)

Syp. Lactulose- 15ml ODHSxcont.

Inj. Pantoprazole(40) IV stat then ODACx Cont.

Tab. Domperidone(10) 1tab stat then TDAC xcont.

Inj. Levetiracetam(500) iv stat then iv TDSx cont. (If h/o Convulsion present)

Other Comorbidities.(Hypothyroid, DIABETES, or anything else).

IV FLUID – not needed

You can use Inf. 20 % Mannitol 10ml IV TIDx 5days if infarction is large(rougy around 50% of cortical lobe)

Hemorrhagic CVA

RT/CATHETER if unconscious and can't take food orally. High risk Consent from patient party. **Never give aspirin to Hemorrhagic cva patients

Adv:

Railed cot/ Air mattress or water bed.

Frequent changing of posture to be encouraged.

Diet: S/R Diabetic (a/c to comorbidities) diet.

Moist O2 inhalation @6-8 L/min with face mask.

Inf. 20% Mannitol -100ml stat then IV TID × 3-5 days.

Inj. Pantoprazole 40 iv stat then iv odac× cont.

Tab. Domperidone(10mg) 1 tab TID× cont

Syp. Lactulose- 15 ml ODHS × Cont.

Tab. Amlodipine (5mg) 1 tab OD× cont.

Tab. Atorvastatin(10mg) 1 tab ODHS× cont.

Inj./Tab Levetiracetam(500mg) 1 tab or 1 amp iv BD×Cont.

Monitor vitals.BP, CBG 6 hourly.

Inform SOS

If BP >220/110 give labetalol (20mg) slow iv

OP(ORGANOPHOSPHATE) POISONING

Remove all the contaminated Clothes

Gastric Lavage....

Most effective within 12 hr but we should do anytime patient appear as it is secreted from gastric mucosa.

History. IPC. High risk Consent.

Whether lavage done or not. If present within 4 hr. Then lavage.

Look for pupillary constriction.

1. NPM

2. RT (continuous)+ CATHETER

3. Moist O2-

4. Inj. Atropine (0.6mg/amp) – 10 amp stat....check for signs of atropinization
repeat 10 amp. Every 10 min untill signs appears.

Then, Maintenance dose: 10 amp (=6mg) in each bottle of IVF 8 hourly.

5. IVF(intravenous fluid) DNS 8 hourly (if non Diabetic) or NS 8 hourly.

6. Inj. Pralidoxime (2 gm= 4 amp) iv stat in 100 ml NS over 20 min.

Then,

May be repeated after 1hr.

7. Pantoprazole

8. Ondem

9. Diazepam(5mg/ml) 1 ml i.m. sos.

Study:

****AGEING**

****INTERMEDIATE SYNDROME & DELAYED POLYNEUROPATHY.**

HANGING

- Stabilize the neck first with Philadelphia Collar.
- Inj. Methylprednisolone (30mg/kg in first 1 hr then 5.4 mg/kg/hr for next 23 hr).[To prevent Acute Spinal Cord edema]
- Secure Airway: Moist O2 if SpO2 drops
- Railed Cot.
- Inj. Levetiracetam(500) IV BD .[Not phenytoin as it is enzyme inducer]
- RT+ Catheter.
- Abx: Inj. Ceftriaxone(1g) iv bd
- Pan
- Ondem

Neurosurgery refer. CCU refer.

MENINGITIS MANAGEMENT

Inj. Ceftriaxone(2g) iv stat then iv BD× cont(AST)

Inj Vancomycin(1g) in 100ml NS to be transfused over 30 min IV BD× cont.(AST)

Inj. Acyclovir(500mg) in 100ml NS IV TID× 14 days(AST)

Inj. Levetiracetam(500) iv bd

Tab. Doxycycline (100mg) 1 tab BD

Syp lactulose- 30ml odhs.

Inj. Dexamethasone (6mg) iv BD× cont.[Bacterial meningitis]

Antimalarial: Artesunate(60) - 2.4mg/kg BW. at 0,12,24 hrs then OD.

MALARIA MANAGEMENT

Vivax:

Chloroquine(250mg with 150mg base) – 4 -4-2 tab D1,D2,D3.

Tab. Primaquine(15mg) 1 tab OD× 14 days[G6PD to be done in male]

If Chloroquine resistant,

Inj. Artesunate (2.4mg/kg)

✓ each vial contain 60mg.

✓ To be diluted with 1ml of 2.5 or 5% NaHCO₃ buffer as provided with the kit
PLUS NS 9ml= Total 10ml solution

✓To be given IV slowly(over 2-5 min)

✓ To be repeated at 12 hr and at 24 hr.

✓ Then OD × cont.

Then, Start oral ACT: Tab. Lumerax 80mg- 1 tab BD× 3 days.

Falciparum Malaria:

Uncomplicated

Tab. ACT-SP[artesunate- SP]

D1: AS 1 + 2SP

D2: AS 1 + PQ(45mg) single dose

D3: AS1

Or,

Lumerax 80 1 tab BD × 3days.

In Pregnancy , no PQ(Primaquine)

1st TM

Tab. Quinine salt(10mg/kg) TDS× 7 days[To be taken with food d/t recurrent hypoglycemia], ECG monitoring.

2nd and 3rd TM

Tab. ACT-SP as earlier.

Complicated

Inj. Artesunate (2.4 mg/Kg) as earlier f/b Tab. ACT-SP or ACT-AL ×3 days.

In pregnancy,

Inj. Quinine Salt 20 mg/kg IV/IM stat f/b 10 mg/kg TDS× 10 days

Mixed vivax+ falciparum:

ACT for 3 days(like falciparum Malaria) and PQ (15mg) for 14 days.

DCLD WITH ASCITES(?Anasarca under Evaluation)

H/O: alcoholic, previously diagnosed or not, Known HEART DS/ KIDNEY DS(urine output) ., COMORBIDITIES. Previous ascitic tap, colour of the fluid. Any diagnosis. Any Blood transfusion. Hematemesis/Melaena/Jaundice/ Infection(UTI, Pneumonia)

? PAIN ABD WITH ASCITES. (? SBP)

Inv:

BP, CBG must.

Urgent Na, K, Ur, cr

EDTA + Clot(LFT, RFT, Amylase, lipase if pain abd.)

PT, aPTT, INR.

ASCITIC FLUID STUDY.

USG W/A(if not diagnosed as CLD)

Urine R/E, M/E – to exclude UTI which can ppt HEPATIC ENCEPHALOPATHY)

CXR- PA view.

Serology(for UGI endoscopy as well as CLD cause hepB, C)

Advice:

1. S/R high protein diet.
2. Moist O2 if distress, @2-4 L/min with face mask.
3. Decubitus: propped up. If distress present.
4. Inj. Lasix(20) iv BD× Cont.
5. Tab. Aldactone[Spironolactone](100) 1 tab OD× cont. [Lasix: Aldactone= 2:5]

6. Syp. LACTULOSE – 15 ml TDS× Cont. till loose stool occur then decrease frequency.

[****VVI DRUG, SEND PARTY TO BUY ON ADMISSION]

7. Inj. Pantoprazole(40) – iv odac× cont

8. Tab. Domperidone(10) 1 tab TDS× Cont.

9. Inj. Cefotaxime(1g) iv TID × Cont. (if h/o previous ascitic tap, pain abd and fever, for prophylaxis against SBP).

10. Drugs for comorbidities.

Checklist:

Abdominal girth, Colour of stool, Malaena/Hematemesis, LFT, U/O, Oedema, PT, aPTT, INR. Sleep Pattern.

Hepatic Encephalopathy

[Look for hepatic flap,GCS]

Inv: NCCT Brain, E+C, CXR-PA, Ascitic fluid diagnostic tap, Urine R/M ,C/S, BP,CBG.

Advice:

RT, Catheter.

Railed cot

Water bed/Air mattress

- RT feeding 100ml 4 hourly× cont (Salt restricted)
- Moist O2 SOS
- SYP. LACTULOSE- 30ml q4hourly× cont [2-3 loose stools/day]
- Tab. Rifaximin(550) 1 tab BD× Cont.
- Inj. Pan 40 IV OD× cont.
- Tab. Domperidone (10) 1 tab TDS × Cont

- Tab. Propranolol (40) ½ tab BD× cont(if ongoing bleeding, inj. Terlipressin(1mg) 1 amp IV TDS× 3 days).
- Antibiotics: Inj. Cefotaxime(2g) 1amp IV stat f/b IV TDS× cont(AST).
- MVI – 1 amp in 100ml NS IV OD over 15 min× cont.
- NO DIURETIC.(CONTRAINDICATED)
- CBG monitoring 12 hourly.
- If malaena occurs, 2 unit PRBC requisition urgently.

Don't Hesitate to ask, after all, questioning is the 1st step to learning. 😊

Dr Subhajit Mondal, GM, U-3, NRSMCH.

Anemia under Evaluation

1.Urgent CBC.

2.EDTA(comete hemogram) +CLOT(LFT, RFT, LDH,)

Extra EDTA, CLOT. ? (At least 2)

3.Hemato refer(for cbc ps, reti count, dct)

-CBC-PS with RETI COUNT

--DCT,

4.ANA, ENA PROFILE, C3, C4

5.-Fe profile,

----S. VIT-B12, FOLIC ACID.

6.CXR- PA view.(cardiomegaly)

7.? Stool OBT, OPC(if normocytic, normochromic, or Microcytic hypochromic)

8.2unit PRBC requisition

Plan later:

BMA + BM-BIOPSY.

Snake bite

Inj. AVS – 10 vial in 500ml NS IV over 30 min stat.

Inj. TT(0.5mg) 1 amp IM stat

Inj. Co-amoxycylav(1.2gm) IV stat then IV TDS× cont.(AST)

Inj. Metrogyl (500) IV stat and IV TDS× Cont.(AST)

Inj. Pan 40 iv odac × cont.

Inj. Ondem (4) 1 aml sos.

Inj.Adrenaline(1:1000) 0.5 ml s/c stat if anaphylactic reaction occurs.

Blood for urgent cbc, na,k,ur,cr, Urine R/M.

E+C.

20WBCT and SBC(Single Breath Count) to be mentioned.

Hematological Malignancy(AML/CML/CLL/LYMPHOMA/MM)

Diet: normal (a/c to comorbidity)

Moist O2 @ 4l/min with face mask (as per SpO2)

RT/Catheter (if altered sensorium) ;

PRBC/PLATELET requisition

- IV Fluid: NS 500ml 6 hourly
- Antibiotics: Broad spectrum- Inj. Pipzo(4.5 g) IV stat f/b IV TDS× cont (AST)
Tab. Voriconazole(200) 2 tab stat f/b 1 tab OD× cont.
May consider Amikacin(500) BD
- Tab. Allopurinol (300) 1 tab OD× cont
- Inj Pan40 iv odac× cont
- Inj. Ondem(4) IV BD× cont

Investigation: E+C, HEMATO refer- CBC-PS, BM Biopsy, USG W/A,CXR-PA(Digital), [Urine R/M,C/S,Blood C/S is f/o septicemia]

AECOPD

When to suspect?

Known chronic smoker, with sudden increase in cough, dyspnea, wheezing, or change in amount and character of sputum. And a history of previous exacerbation strongly suggestive. On examination, diffuse crepts, rhonchi+/-, barrel shaped chest(AP diameter = Transverse diameter), accessory respiratory muscles working, Tachypnoea,cyanosis(central). On Chest X-Ray: Hyperinflated lung field, emphysematous bullae may be visible, prominent pulmonary trunk, RVH if chronic pulmonary hypertension persists.

Advice:

Cessation of smoking.

Moist O2 @6-8 L/min or more as per spo2[Target SpO2>= 90%]

Diet: Normal diet or as per comorbid condition.

Neb(sNebulization) with DUOLIN- 1 respule 6 hourly (each session at least for 20min) × cont

Neb(s. With BUDECORT- 1respule 12 hourly× cont.

Antibiotic: Inj Ceftriaxone(1g) IV BD + Azithromycin (500) 1 tab ODAC× 5 days or Pipzo(4.5gm) IV TDS.(APST)

Steroid: Inj. Hydrocortisone (200) IV Stat f/b 50mg IV QDS× cont.

IV fluid: as per volume status.

Syp Lactulose- 15 ml ODHS× cont

Tab. N-acetyl cysteine (600) 1 tab BD× cont

Tab. Multivitamin- 1 tab OD× 2weeks

Inj. Pantoprazole(40) IV ODAC × cont

Inj. Ondem(4) IV TDS× cont.

Tab. Paracetamol(500) 1 tab SOS.

Inj. Deriphylline- 2 amp IV TDS × cont.(If bronchospasm still persists) [nausea, tachycardia,tremor are side effects]

Vaccine: Influenza vaccine annually/Pneumococcal vaccine/ vaccination for Bordetella pertussis (foreign countries)

Non-invasive Positive pressure ventilation (NIPPV) – is indicated in patients with respiratory failure (mainly Type 2) define as $\text{PaCO}_2 > 45 \text{ mmHg}$. Contraindication: CVS instability, altered sensorium, uncooperative patient, Craniofacial abnormality precluding effective fitting of mask and unable to clear copious secretion.

INVESTIGATION:

E+C

CXR-PA(Digital)

Sputum for gram stain,C/S, AFB, Fungal Stain

12lead ECG [To r/o cor pulmonale or electrolyte imbalance, Pulmonary embolism]

2D ECHO

ABG

CT Thorax(next line of investigation)

ABG

Normal Values:

PaO_2 - 90-100 mmHg,[SpO_2 - 90% means PaO_2 -60mmHg. Then every 1% fall in spo_2 = 1.5 mmHg fall in PaO_2 . i. e at 80% Spo_2 , PaO_2 will be 45mmHg.]

PaCo_2 - 35-40 (40) mmHg

HCO_3 - 22-26(24) mEq/L

pH: 7.35-7.45

pH < 7.35 acidosis and vice versa.

Step 1: Look at PaCO₂ and pH, if anyone is out of range then it is acid base d/o.

Step 2: If both is abnormal,

if both change in same direction, it is primary metabolic acid base d/o.

if both change in opposite direction then, it is primary Respiratory acid base d/o.

Step 3: If either pH or PaCO₂ abnormal then, the condition is mixed metabolic and respiratory disorder (i.e equal and opposite)

if only PaCO₂ is abnormal, then directional change in PaCO₂ will determine the type of respiratory d/o. [high PaCO₂ means respiratory acidosis] and opposite metabolic d/o [with metabolic alkalosis]

if only pH is abnormal, then directional change in pH identifies the type of metabolic disorder [low pH indicates a metabolic acidosis] and the opposite respiratory d/o. [here, respiratory alkalosis].

Step 4: For primary Respiratory d/o,

Respiratory acidosis:

Acute: for every 10mmHg increase in CO₂, HCO₃ increase by 1meq

Chronic: For every 10mmHg decrease in CO₂, HCO₃ increase by 4meq

Respiratory Alkalosis:

Acute: for every 10mmHg decrease in CO₂, HCO₃ decrease by 2meq

Chronic: for every 10mmHg decrease in CO₂, HCO₃ decrease by 4meq.

Eg:

Acute Foreign body ingestion, pH: 7.33, PaCO₂- 70, HCO₃- 27,

Then?

Acute condition : pH and PaCO₂ opposite direction so, Respiratory d/o

Δ PaCO₂= 30. So, Δ HCO₃= 3×1=3meq

HCO₃ should be 26+3= 29. So, compensated Acute Respiratory Acidosis with metabolic alkalosis.

Step 5: For primary Metabolic disorder,

If metabolic acidosis:

Expected PaCO_2 : $40 - [1.2 \times (24 - \text{HCO}_3) \text{ or } \Delta\text{HCO}_3]$

If metabolic alkalosis:

Expected PaCO_2 : $40 + [0.7 \times \Delta\text{HCO}_3 \text{ OR } [\text{HCO}_3 - 24]]$.

.....

If expected value matches with the given value, then it is compensated, otherwise uncompensated.

Anion Gap [Δ/Δ gap]

$\text{AG} = \text{Na} - [\text{Cl} + \text{HCO}_3] = [\text{UA} - \text{UC}]$

Normal value = 8-12 meq/L. (std=12)

$\text{AGc} = \text{AG} + 2.5 \times [4.5 - \text{albumin}]$

In case of High anion gap metabolic acidosis,

Δ/Δ gap or GAP-GAP RATIO :

$\text{AG Excess} / \text{HCO}_3 \text{ deficit} = [\text{AG} - 12] / [24 - \text{HCO}_3]$

Δ/Δ gap:

= 1, high anion gap metabolic acidosis with no secondary acidosis.

>1 high anion gap metabolic acidosis with coexisting metabolic alkalosis. (HCO_3 loss is lesser than expected or Δ anion gap)

<1 high anion gap metabolic acidosis with secondary metabolic acidosis (as HCO_3 loss is greater than expected or Δ Anion gap)

NB: These all are applicable only in High Anion Gap Metabolic Acidosis

Dr. Subhajit Mondal. U-3 GM. NRSMCH.

I will explain every bit in detail. 😊

Any Unconscious patient: 6 thumb rule

1. Pulse
2. BP
3. CBG
4. DRUG history: Antihypertensive (Thiazide diuretic), Hallucinating/Antipsychotic/Sedative
5. CNS Examination: Planter, Neck rigidity with Fever, any focal neurological deficit (Hemiparesis etc)/ h/o TRAUMA.
6. ? Known CKD.(Uremic Encephalopathy)/DCLD/HYPOTHYROIDISM(myxoedema Crisis).